

1 The e.m.f. induced in a coil by a changing magnetic flux is equal to the rate of change of flux with time. Which is a unit for magnetic flux?

A $\text{kg m}^2 \text{s}^{-2} \text{A}^{-1}$

B $\text{kg m}^2 \text{s}^{-2} \text{A}$

C $\text{kg m}^2 \text{s}^2 \text{A}^{-1}$

D $\text{m}^2 \text{s}^{-2} \text{A}^{-1}$

Ans: A

Using Faraday's Law,

$$E = - \frac{d\Phi}{dt}$$

$$\frac{P}{I} = - \frac{d\Phi}{dt}$$

$$\frac{P}{I} \Delta t = -\Delta\Phi$$

$$\text{Unit for } \Phi = \frac{\text{kg m}^2 \text{s}^{-2} \text{s}^{-1}}{\text{A}} \text{s} = \text{kg m}^2 \text{s}^{-2} \text{A}^{-1}$$